

Lab 4: Deploying a Web Application on EC2

Duration: 30 minutes · **Region:** us-east-2 (Ohio)

Objectives

- Install a web server and deploy a page on your instance
 - Create an AMI from it and launch a second identical server in another AZ
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Exercise 4.1: Deploy on web-1 (10 min)

In the Instance Connect terminal for `userN-web-1` :

```
sudo dnf install -y httpd
sudo systemctl enable --now httpd
echo "<h1>Hello from $(ec2-metadata -i | cut -d' ' -f2) in $(ec2-metadata -z | cut -d' ' -f2)</h1>" | sudo tee /var/www/html/index.html
```

In the EC2 console copy the **Public IPv4 address** of `userN-web-1` and open `http://<public-ip>` in a new tab (http, not https). You should see the instance ID and `us-east-2a` .

Page not loading? In order: instance Running → security group allows HTTP 80 → you typed `http://` → `sudo systemctl status httpd` shows active.

Exercise 4.2: Create an AMI (5 min)

1. Select `userN-web-1` → **Actions** → **Image and templates** → **Create image**.
2. Image name `userN-web-ami` → **Create image**.
3. **AMIs** (left menu) — it shows *Pending* for 2–4 minutes, then *Available*. Read ahead while you wait.

Exercise 4.3: Launch web-2 from the AMI (15 min)

The AMI has Apache and the page baked in, but the page still names web-1. A **user-data** script runs at first boot to fix that automatically.

1. **AMIs** → select `userN-web-ami` (Available) → **Launch instance from AMI**.
2. Same settings as Lab 3, except:

Setting	Value
Name	userN-web-2
Subnet	userN-subnet-public2-us-east-2b
Auto-assign public IP	Enable
Security group	existing → userN-web-sg
IAM instance profile	userN-ec2-role
Advanced details → User data	paste below

```
#!/bin/bash
echo "<h1>Hello from $(ec2-metadata -i | cut -d' ' -f2) in $(ec2-metadata -z | cut -d' ' -f2)</h1>" > /var/www/html/index.html
```

3. **Launch instance.** When Running, browse to **its** public IP. Different instance ID, **us-east-2b** — and you never logged in.

Manual install → AMI → user data is how one server becomes a fleet. A load balancer in front of both is the next step (lecture).

✓ Keep running

userN-web-1 , userN-web-2 , userN-web-ami — cleaned up in Lab 8.

🧠 Knowledge Check

1. What does an AMI capture that user data does not?
2. Why put web-2 in a different Availability Zone?